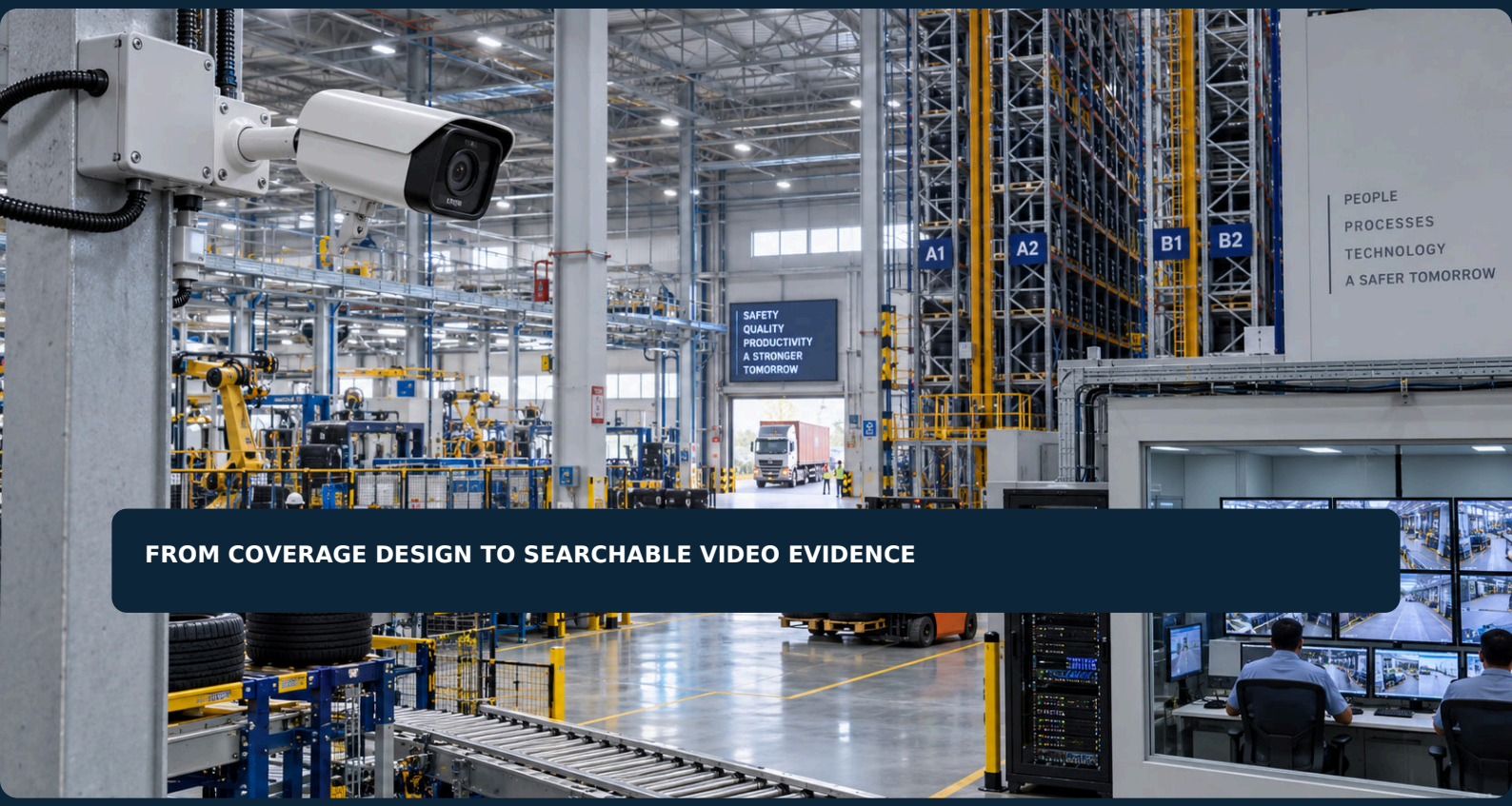
INDUSTRIAL IP VIDEO SURVEILLANCE / MRF DAHEJ

CCTV Solution
Design, Deployment & Service

Plant-wide visibility. ASRS monitoring. Structured lifecycle support.



FROM COVERAGE DESIGN TO SEARCHABLE VIDEO EVIDENCE

87	2	8 MP / 5 MP	~USD 0.029M
DOCUMENTED CAMERAS	INTEGRATED PROJECTS	MIXED-USE DESIGN	ORDER VALUE

CLIENT: MRF LIMITED, DAHEJ / SOLUTION INTEGRATION: D-TECH

Conceptual industrial surveillance visual for communication. Documented scope is based on the awarded orders.

EXECUTIVE SUCCESS STORY

Two CCTV projects. One scalable foundation.

D-TECH supported MRF Dahej with a plant-wide 8 MP deployment and a focused ASRS/dip-roll system, combining cameras, PoE, recording, storage, cabling, mounting and monitoring.

72	15	3	14
8 MP PLANT CAMERAS	5 MP ASRS CAMERAS	NETWORK VIDEO RECORDERS	SURVEILLANCE DRIVES

BUSINESS NEED

Reliable video coverage for operations, assets and evidence.

Wide plant areas required high-resolution coverage; the ASRS environment needed focused views, a dedicated monitoring point, protected hardware and a documented 30-day recording basis.

END-TO-END DELIVERY

01	02	03	04
DESIGNED	DEPLOYED	TESTED	SUPPORTED
Coverage, placement, bandwidth and storage.	Cameras, PoE, NVRs, cabling, racks and displays.	Image quality, recording, playback and retention.	Warranty, fault response and lifecycle service.

OUTCOME A modular system for live visibility, incident review, operational control and future expansion.

SOLUTION DESIGN METHODOLOGY

Camera counts follow risk, scene and evidence objectives.

The design process converts operational needs into camera views, network capacity, retention and physical infrastructure.

01 Operational survey

Identify production, ASRS, loading, movement and critical equipment scenes.

02 Scene objective

Define overview, recognition, identification and incident-evidence needs.

03 Lens and mounting

Select field of view, IR range, height, bracket and environmental protection.

04 Network design

Map PoE ports, uplinks, cable length, switching capacity and rack locations.

05 Storage sizing

Align resolution, frame rate, codec, motion profile and retention.

06 Control-room design

Plan NVR, displays, operator view, playback and export workflow.

07 Installation engineering

Plan cable trays, conduit, back boxes, stands, earthing and labels.

08 Acceptance plan

Define image, recording, search, backup, retention and handover tests.

DESIGN PRINCIPLE

Every camera should answer an operational question and produce usable evidence, rather than simply add another live view.

CCTV SOLUTION ARCHITECTURE

From industrial scene to secure video evidence.

The architecture separates field capture, power, transport, recording, storage and operator use, with a clear path for health monitoring.

F **FIELD CAMERAS**

8 MP plant and 5 MP ASRS IP cameras

E **EDGE POWER**

PoE switching and protected termination

N **NETWORK**

Cat6, uplinks, trays and conduit

R **RECORDING**

64-channel and 32-channel NVRs

S **STORAGE**

Surveillance drives and retention policy

O **OPERATIONS**

Display, live view, playback and export

SECURITY OPERATIONS LOOP

Live monitor → alert or observation → playback → export → action. Labels, synchronized time and retention checks make the footage useful during investigations.

SERVICE VISIBILITY Camera, PoE, recorder, disk and display status belong in preventive maintenance.

PLANT-WIDE SURVEILLANCE

72-camera 8 MP plant surveillance platform.

The first documented deployment created a broad IP-video layer with high-resolution cameras, centralized recording, PoE access and extensive outdoor cabling.

SUBSYSTEM	DOCUMENTED SCALE	PURPOSE
IP cameras	72 × 8 MP AcuSense fixed bullet cameras	Plant coverage
Recording	2 × 64-channel NVRs, 8 SATA bays each	Playback
Storage	12 surveillance hard-disk drives	Evidence capacity
PoE network	6 managed PoE switches	Power and data
Cabling	Approx. 5.5 km outdoor Cat6	Field connectivity
Field hardware	72 stands / brackets and accessories	Stable mounting
Monitoring	Commercial display and HDMI connectivity	Live view
Warranty	Camera and system component coverage	Lifecycle

DESIGN VALUE

High pixel density supports stronger identification and post-event review.

Focus: coverage consistency / outdoor cabling / recording capacity / serviceable mounting

ASRS AND DIP-ROLL MONITORING

15-camera system for automated warehouse operations.

The second order extends coverage into ASRS/dip-roll operations with a dedicated recorder, 30-day storage basis, managed PoE, rack equipment and protected pathways.

SUBSYSTEM	DOCUMENTED SCALE	PURPOSE
IP cameras	15 × 5 MP bullets with 50 m IR	Aisle visibility
Recording	1 × 32-channel NVR, 4 SATA interfaces	Dedicated video
Retention	2 × 10 TB surveillance drives	30-day basis
PoE switch	1 × managed 24-port Gigabit PoE	Aggregation
Rack	17U floor-mount network rack	Serviceability
Display	50-inch 24×7 commercial display	Monitoring
Pathways	400 m conduit + 100 m covered GI tray	Protected routing
Mounting	15 back boxes + 15 fabricated stands	Field quality

DESIGN VALUE

Dedicated ASRS recording keeps operational evidence organized and accessible.

Focus: IR performance / aisle coverage / retention / protected pathways / rack discipline

PHYSICAL DEPLOYMENT ENGINEERING

Infrastructure quality determines reliability.

Physical design keeps cameras powered, connected, protected and maintainable in an industrial environment.

01 Engineered mounting

Camera height, angle, vibration control and fabricated stands.

02 Weather protection

IP65 back boxes, sealed entries, conduit and flexible hose.

03 Cable pathways

Covered GI trays, supports, bends, couplers and routing.

04 Structured cabling

Cat6 termination, patch panels, IOs, patch cords and RJ45s.

05 Rack discipline

17U rack, PDU, fans, tray, cable management and labelling.

06 PoE availability

Port planning, power budget, uplinks and spare headroom.

07 Display readiness

Commercial display mounting, HDMI and operator visibility.

08 Safety execution

MRF EHS compliance, work permits, PPE and controlled work.

MAINTAINABILITY

Clear labelling, accessible junctions and organized racks reduce fault-isolation time and support safer changes.

COMMISSIONING AND ACCEPTANCE

Every view, path and recording function is verified.

Commissioning turns installed hardware into an operational system with documented settings, validated retention and a handover baseline.

01 Physical inspection

Mounting, enclosures, routes, labelling and connector quality.

02 Network test

IP plan, PoE, link stability, bandwidth and packet continuity.

03 Camera tuning

Focus, field of view, exposure, IR, time and framing.

04 Recording test

Recording, playback, search and export verification.

05 Storage check

Disk health, overwrite behaviour and retention observation.

06 Display test

Live layouts, alarm visibility and operator workflow.

07 Failure simulation

Camera/link interruption, recorder restart and recovery.

08 Handover

As-built list, baseline settings, training and acceptance record.

ACCEPTANCE STANDARD

A camera is accepted when its scene is useful, recording is searchable and the evidence path is operational.

SERVICE AND LIFECYCLE SUPPORT

Preventive care plus structured incident response.

CCTV retains value when cameras remain online, storage remains healthy and operators can retrieve the needed evidence.

HEALTH CHECK	Camera, PoE, NVR, disk and display status.
PREVENTIVE	Cleaning, focus, IR, connectors and rack review.
INCIDENT	Remote triage, site response and fault isolation.
RESTORE	Repair, replace, reconfigure and validate.
REPORT	Cause, action, closure evidence and recommendation.

SERVICE KPIs

Camera uptime / recording availability / disk health / image quality / response / closure time

CUSTOMER VALUE AND EVIDENCE

From live visibility to defensible incident evidence.

The combined foundation supports security, operations, EHS review and faster investigations across plant and warehouse environments.

01 Broader visibility

87 documented camera positions cover key operational scenes.

02 Stronger evidence

8 MP and 5 MP imaging improves detail for review.

03 Faster investigation

Recording, playback and export reduce search effort.

04 ASRS accountability

Dedicated cameras and retention support process review.

05 Maintainable network

Managed PoE, cabling and racks simplify support.

06 Retention confidence

Storage is sized and tested against requirements.

07 Safer field execution

Mounts and protected pathways align with EHS needs.

08 Expansion readiness

Modular IP architecture supports controlled growth.

DOCUMENTED PROJECT EVIDENCE

PO 4300107880 (21 Apr 2025) + PO 4300118222 (07 Jul 2026). Combined order value is presented as approximately USD 0.029M; detailed rates, contacts and signatures are excluded.

D-TECH: DESIGN / SUPPLY / INSTALLATION / COMMISSIONING / SERVICE

REPEATABLE BLUEPRINT

A repeatable industrial CCTV blueprint.

A reusable delivery approach for manufacturing, chemicals, pharma, energy, logistics, ports, campuses and critical infrastructure.

MRF LIMITED / DAHEJ

Every critical scene visible.
Every event searchable.
Every system serviceable.

D-TECH integrates surveillance design, networking, recording, physical deployment and lifecycle service into an accountable solution.

D-TECH CCTV CAPABILITY

Industrial surveillance from survey to support.

- MANUFACTURING
- WAREHOUSE
- CHEMICALS
- PHARMA
- ENERGY
- LOGISTICS

DESIGNED FOR COVERAGE / DEPLOYED FOR RELIABILITY / SUPPORTED FOR CONTINUITY